

Managing Grating Parameters for Longitudinal Steering with 1D Optical Phased Arrays

Nam-Hyun Kwon, Hyeonho Yoon, Seong-Hwan Kim, Jae-Yong Kim, Junhyeong Kim, Jinhyeong Yoon, Jong-Bum You, Hamza Kurt, and Hyo-Hoon Park

Abstract— We propose and experimentally validate an efficient solution to extend the longitudinal beam steering range, managing all available diffraction parameters in the grating radiators of optical phased arrays (OPAs). Using the thermo-optic tunable radiator, varying the effective index of the grating radiator, the beam was continuously steered to a limited range. The operating wavelength was discretely changed with different wavelength sources to extend and multiply the steering range. Also, the incident direction of light was changed utilizing the bidirectional radiators shared with two subarrays of OPA to extend the steering range and mirror symmetrically. From 1×64 bidirectional OPAs consisting of p - i - n electro-optic phase shifters and MZI path switches to change the incident direction, seamless and continuous longitudinal steering was demonstrated in a range of $\pm 6^\circ$ for a transversal steering range of $\pm 22.7^\circ$ with a divergence angle of 0.5° and 1.0° in each direction.

Index Terms—Optical phased array, Bidirectional radiator, Tunable radiator, p - i - n phase shifter, Silicon photonics,

I. INTRODUCTION

The optical phased array (OPA) has been attracting attention as a component for light detection and ranging (LiDAR) systems, and free space communication, due to the advantages of solid-state beam steering and CMOS-compatible integration [1], [2], [3]. Compared with the existing commercially available mechanical scanning LiDAR systems, it provides high-density integration, high durability, and scalability [4].

Most LiDAR applications require two-dimensional (2D) beam steering with a wide steering range. The 1D phased array can only steer the beam in the transversal direction at a fixed wavelength. Therefore, $M \times N$ 2D phased arrays have been studied to enable 2D beam steering [4], [5]. However, achieving a wide steering range with a 2D phased array is challenging due to special constraints imposed by the placement of many components such as waveguides, phase shifters, and radiators in a unit pixel. However, 1D OPA has less special constraint between radiators and it is worth implementing 2D scanning with 1D OPAs for LiDAR applications.

In order to add a longitudinal beam steering function to the 1D OPA, several methods have been implemented, based on the

principle of diffraction in the grating radiator. The wavelength-controlled longitudinal beam steering method is commonly used [6], [7], [8], [9]. However, it requires a tunable laser source for wavelength tuning, and the tuning range is limited due to the bandwidth of the tunable laser and other components, such as the input grating coupler and power splitters. Instead of a tunable laser, two discrete lasers have been used, employing MZI switch to select a laser source and to change the input wavelength for the same OPA [10].

As an alternative method, longitudinal beam steering by thermal tuning of the refractive index of a grating radiator has been studied [11], [12], [13]. Thermal tuning of the grating radiator requires heating of silicon or insulator waveguide up to several hundred $^\circ\text{C}$. To control the temperature of the grating radiator located on the silicon layer, metal heaters are placed above the radiator [4], [12], or impurity-doped semiconductor heaters are fabricated in the silicon radiator [1], [11], [12]. In this method, the longitudinal steering range is restricted by the allowable heating temperature, and it becomes narrow down when the scale of the radiator increases to achieve a narrow divergence angle with a wide grating aperture since the heating power increases almost proportionally with the heating area. As another method, the integration of many 1D OPAs having different grating periods [14] was examined. In this scheme, a couple of OPAs are mirror-symmetrically arranged sharing the same bidirectional radiators to extend the longitudinal steering range to the opposite direction through changing incident direction of the light. The 2D configuration of 1D OPAs requires a large integration area and cannot provide continuous steering in the longitudinal direction.

In this paper, we studied a solution, combining the wavelength change, bidirectional input, and thermo-optic tuning of the grating, to extend the longitudinal radiation range based on the 1D OPA. The steering range can be extended and multiplied for each trial to the wavelength and incident direction, and the operation of tunable radiator links with continuous steering between the discrete changes. Through the appropriate design of the diffraction parameters, we experimentally demonstrate a seamless and continuous beam steering function.

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ksh113@kaist.ac.kr, jaeyong21.kim@kaist.ac.kr, wnsud@kaist.ac.kr, ymyyjh@kaist.ac.kr, hamzakurt@kaist.ac.kr, parkhh@kaist.ac.kr). J.-B. You is with the National Nanofab Center (NNFC), Daejeon, Korea (e-mail: jbyou@nnfc.re.kr)

II. DESIGN AND SIMULATION

Figure 1(a) shows the schematic of designed bidirectional 2×64 optical phased arrays that share the input light source and tunable grating radiators. It consists of an input grating coupler, 1×2 Mach-Zehnder interferometer (MZI) switch, multi-mode interferometer (MMI) power splitters, p - i - n phase shifter arrays, and tunable grating radiators. The devices were integrated on the silicon-on-insulator (SOI) substrate. The input grating coupler was designed to incident two wavelengths of 1525 nm and 1550 nm. It is linked with a thermo-optic MZI switch to select one part of the bidirectional OPAs. In each OPA part, multi-stages of 1×2 MMI splitters, the p - i - n phase shifters, the phase-feeding lines, and the grating radiators were configured similarly to previous work [11], [15]. The two OPA parts (subarrays) were configured with a mirror symmetry, possessing commonly the grating radiators placed at the center. The splitters and phase shifters are identically designed in the two subarrays.

The tunable radiator was designed based on the diffraction theory of the grating. Radiation angle from the grating can be expressed as [3]:

$$\theta_{\perp} = \sin^{-1}(n_{\text{eff}} - \lambda/\Lambda) \quad (1)$$

where $\theta_{\perp}$ is the longitudinal radiation angle from the normal direction., n_{eff} is the effective index of the grating radiator, λ is the free space wavelength, and Λ is the grating period. In Eq. (1), there are three variables to determine the radiation angle. Among them, the Λ is fixed in the fabricated device and other parameters of the n_{eff} and λ can be varied in device operation. To control the n_{eff} of the radiator, we formed the n - i - n junction heater in the radiator region. The n - i - n heater requires less carrier injection than other junction-based heaters, such as p - n heater and p - i - n heater, and thus it can provide a high resistance with relatively low optical loss [12]. Compared to the metal-ohmic resistors forming on an insulating layer over a silicon (Si) waveguide, the Si resistor can be configured without screening the radiated beam and can increase operation speed by fast heat dissipation through the Si layer. To form the n - i - n junction in the radiator, the entire grating radiator is left as an intrinsic region, and the n -doped regions are placed on both sides of the grating region, as shown in Fig. 1(b). The metal electrodes were formed on the n -doped regions.

The grating radiator array was designed with a waveguide width of $1 \mu\text{m}$ and a spacing between waveguides of $2 \mu\text{m}$ to achieve a transversal steering range over 45° . The grating period Λ was designed to form the diffracted beam toward a normal angle ($\theta_{\perp} = 0^\circ$) at a wavelength of 1550 nm and $\theta_{\perp} = 3^\circ$ at a wavelength of 1525 nm. Using a finite-difference time-domain (FDTD) simulation, the period was determined to be 495 nm with an etch depth of 10 nm and a duty ratio of 0.5. The length of the grating radiator was determined as $300 \mu\text{m}$ to provide a longitudinal divergence angle of less than 0.5° . The width of the 64 grating radiators was $128 \mu\text{m}$.

Figure 1(b) shows how to manage the control parameters for longitudinal beam steering. In the turn-off state of the tunable radiator, the beam is radiated to the normal direction $\theta_{\perp 0}$ at the input wavelength of λ_1 through the subarray OPA 1. The beam is steered to $+\theta_{\perp 1}$ by increasing the heating power of the tunable radiator. At $+\theta_{\perp 1}$, the input wavelength is switched to λ_2 and

then the beam is steered further by operating again the tunable radiator increasing the heating power from zero until reaching $+\theta_{\perp 2}$. To cover the opposite $-\theta_{\perp 1}$ side, the incident light is assigned to subarray OPA 2 and repeat the same process changing the wavelength and operating the tunable radiator as done on the $+\theta_{\perp 1}$ side.

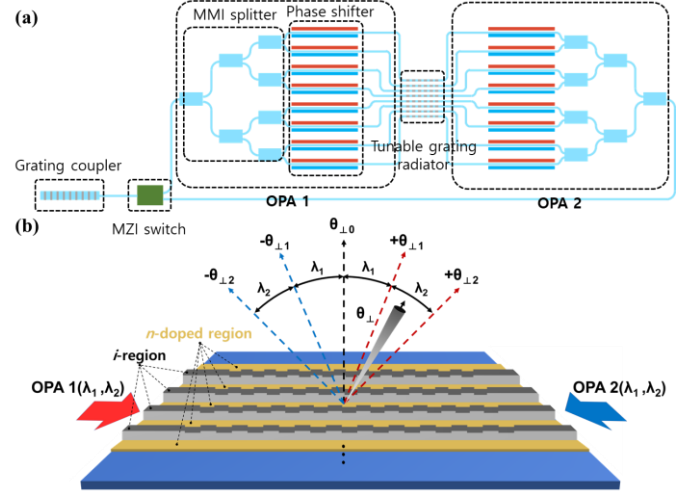


Fig. 1. Schematics of bidirectional OPAs: (a) whole structure consisting of two OPAs shared with the tunable radiator and (b) tunable radiator structure and management of control parameters.

Figure 2 shows the simulation results of the longitudinal beam steering when the tunable radiator is operated. The simulation was performed by a combination of Lumerical DEVICE and FDTD simulations. First, the DEVICE simulation was employed to calculate the temperature distribution resulting from the uniform power applied to the i -region of the whole radiator area of 64-channels. Subsequently, the temperature distribution was transformed to the n_{eff} profile of the grating radiator. Next, the radiation angle of the grating radiator was calculated using FDTD simulation based on the obtained n_{eff} distribution.

The optical fields radiated from the tunable radiators, simulated using FDTD are presented in Fig. 2(a), corresponding to the heating powers varying with 0 W, 3.4 W, and 5.5 W. In the Lumerical DEVICE simulation, the temperature in the grating radiator region reached near 560°C at 5.5W. The longitudinal radiation angle $\theta_{\perp}$ is shifted from 0° to 5.7° , as clearly seen in the Fig. 2(b). The longitudinal beam steering efficiency $\Delta\theta_{\perp}/\Delta P$ becomes about $1.1^\circ/\text{W}$.

III. FABRICATION AND EXPERIMENTAL RESULT

OPA chip was fabricated in a CMOS compatible process using an 8-inch SOI wafer with a 220 nm top silicon layer and a $2 \mu\text{m}$ buried oxide thickness. A KrF scanner with a 248 nm wavelength laser was used for the photolithography. For the ohmic contact, phosphorus was heavily implanted with a dose density of $6.5 \times 10^{15} \text{ cm}^{-2}$ and an acceleration voltage of 30 keV. The secondary ion mass spectrometry (SIMS) analysis indicated that the fabricated n -doped region had a doping concentration of $5 \times 10^{20} \text{ cm}^{-3}$. The aluminum metal electrodes were formed for electrical contact between the chip surface, and

each n-doped region formed between the tunable radiator arrays. The footprint of the fabricated OPA chip was $8.8 \text{ mm} \times 3.3 \text{ mm}$.

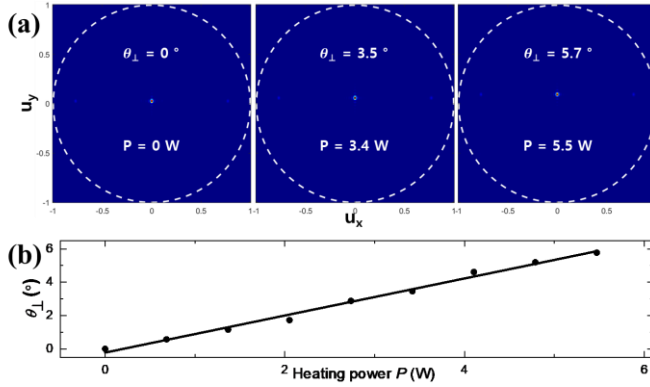


Fig. 2. Simulation results of (a) far-field images when heating power of the tunable radiators is supplied with 0 W, 3.4 W, and 5.5 W and (b) variation of longitudinal radiation angle ($\theta_{\perp}$) with a function of heating power P .

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The performances of the element components of the OPAs were measured from test patterns following the same fabrication processes. The input grating coupler with a 620 nm period showed a coupling loss of less than 5.5 dB at a 1520–1580 nm wavelength range. The MZI 1×2 switch with a 300 nm length of TiN heater showed an optical loss of less than 1 dB and a crosstalk of -26 dB at the first cross-state 5 V. Also, the n - i - n heater fabricated at tunable radiator showed an electrical resistance of 525 Ω .

Figure 3(a) shows a fabricated chip mounted on an evaluation board. The evaluation board is connected to a digital-to-analog converter (DAC) and a power source, enabling it to apply voltage to the components of the OPA and to supply power. For transversal beam steering, the phase shifters were connected to the DAC prepared for voltage control with a high resolution within a range of 0 V to 2 V. The 1×2 switch and n - i - n heater are directly connected to the power source.

The beam steering performance of the fabricated OPA was characterized using the experimental setup described in Fig. 3(b). This setup includes the Fourier imaging system for far-field image measurements and the OPA control system. The radiation direction information of the beam radiated from the OPA was extracted by capturing far-field images focused on the focal plane of the objective lens using an IR camera. A single diffraction beam for each transversal direction was formed through control of the voltages supplied to the phase shifters of the 64-channel optical phased array, employing the hill-climbing algorithm [3], [11]. The phase conditions for steering the single beam were set separately for the subarrays of OPAs 1 and 2.

Figure 4(a) shows the measured far-field images which were obtained by longitudinal steering through combinations of OPA switching, wavelength change, and radiator heating.

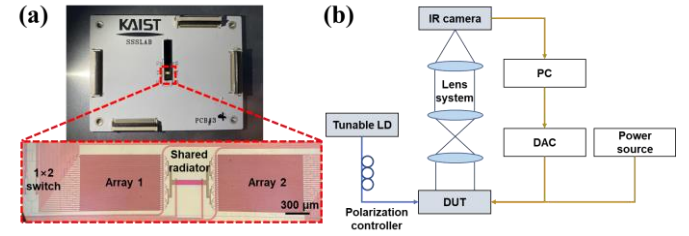


Fig. 3. (a) Photograph of fabricated OPA chip mounted in evaluation board and (b) block diagram of experimental setup to measure beam steering performances.

The full scanned range was from -6° to $+6^\circ$. We note that the single beam condition was maintained during wavelength change and tunable radiator operation, once the beam was formed to the normal direction $\theta_{\perp 0}$ through the phase control of each OPA 1 and 2 at off-state of the radiator. The radiation to the normal direction can be affected by the etching states of the grating. Nevertheless, in the fabricated devices, the off-state beams were well aligned to the $\theta_{\perp 0}$ for both incident directions, showing a similar intensity with neighboring beams, as the simulation (Fig. 2(a)). At the on-state of the tunable radiator, the beam can be continuously steered by increasing heating power, but in Fig. 4(a) the beam was discretely displayed to show the beam position clearly. Note that the beam intensities at 1525 nm were weaker than those at 1550 nm. The result might be due to the different wavelength dependencies of the components, especially, the MMI which was designed for 1550 nm. The steering ranges at each zone could be well matched with the simulated values. However, the electrical input power to heat the radiator was larger than the simulation, as compared in Fig. 4(b). The measured steering efficiency was $0.73^\circ/\text{W}$ which is slightly smaller than the simulated value, $1.1^\circ/\text{W}$, due to some heat dissipation from the radiator region. These efficiencies are much smaller than in the previous work [11], since the heating area of the radiators is enlarged in the present work.

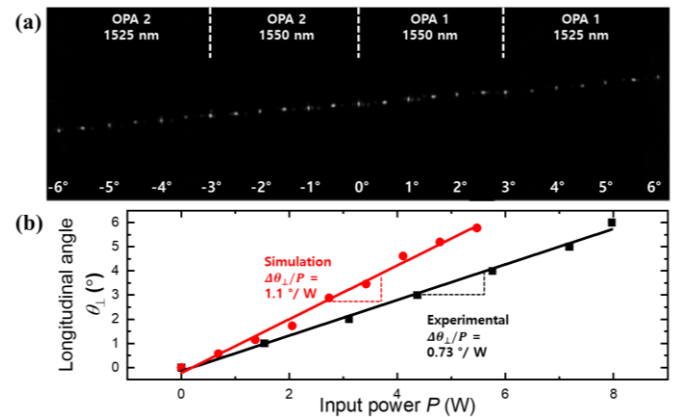


Fig. 4. (a) Measured far-field image of the 2×64 OPA showing longitudinal beam steering by the combination of OPA switching, wavelength change, and radiator heating. (b) Comparison of longitudinal beam steering efficiencies between simulated and experimental results.

In Fig. 4(b), the tunable radiator was operated up to an input power of 8 W to reach the full range of $\theta_{\perp} = 6^\circ$ without wavelength change. However, we only applied up to 4 W to

cover $\Delta\theta_{\perp} = \pm 3^{\circ}$ for both wavelengths. Thus, the longitudinal scanning range can be extended more if the second wavelength is selected as a shorter one and the switching position of wavelength, θ_{11} , is shifted to a larger angle. In the radiator design, to reduce the divergence angle, the area of the radiator should be extended. If the radiator area is increased, high-temperature heating can be a burden in device operation. In such a case, backside etching of the radiator area can be an effective solution to improve thermal dissipation and increase steering efficiency [13]. Note that in Fig. 4 the beam positions of $+\theta_{\perp}$ were symmetrically changed to $-\theta_{\perp}$ with the same heating power. This is an important merit of the bidirectional radiators since it can reduce the efforts to tune the beam position, in addition to saving device area.

Figure 5(a) shows the result of 2D steering, adding the operation for transversal steering. The transversal radiation angle reaches $\theta_{\perp} = -22.7^{\circ}$ and $+22.7^{\circ}$, indicating a maximum transversal steering range of 45.4° . The bias conditions on the phase shifters were readjusted whenever the beam was shifted to a transverse direction and the subarray was changed to flip the longitudinal direction to the opposite side. When longitudinally steering the beam radiating in each transversal direction, it was steered without modifying the bias condition.

Figure 5(b) shows the divergence angle of a typical radiation beam selected from intensity data measured using a far-field imaging system. Measured divergence angles of the in-phased main beam were about 0.5° and 1.0° in transversal and longitudinal direction, respectively. This result is well-matched to the 3D FDTD simulation results estimated as 0.53° and 1.07° for each direction.

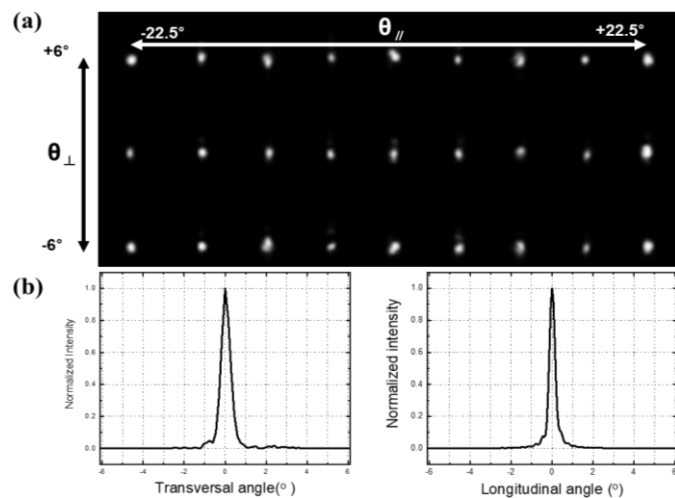


Fig. 5. (a) Far-field images measured during the 2D beam steering with simultaneous operation of transversal and longitudinal beam steering. Field profiles are also presented to show beam divergence angles for (b) transversal and (c) longitudinal direction.

IV. CONCLUSIONS

In conclusion, we have demonstrated a solution to extend the longitudinal beam steering range, utilizing all available diffraction parameters in the grating radiators of OPA: varying the wavelength, refractive index, and incident direction in the

grating radiator. The wavelength was discretely changed with different wavelength sources selected by the MZI switch. The steering range was symmetrically extended using the bidirectional radiators coupled with two identical OPAs. Using the thermo-optic tunable radiator, varying the effective index of the grating radiator, the beam was continuously steered between discretely distributed beam positions determined by changing the wavelength and the incident direction. Successful demonstration of seamless 2D steering through the combinational management of these parameters confirmed the possibility of wide and precise 2D beam scanning in the LiDAR application.

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